

Terms

Read these first. Three are used more narrowly here than in ordinary usage, and every confusing number below traces back to one of them.

- **A lesion** is an area of abnormal tissue, so for us a tumor: one three-dimensional mass a radiologist would put in a report. **In our metrics a lesion** is something much narrower — one connected blob of tumor mask within a *single 2D slice*. A scan is a stack of about 155 cross-sections, so one tumor crossing twenty of them counts as twenty lesions, a thin or broken enhancing rim fragments into several more inside each slice, and a five-pixel speck counts the same as a whole tumor.
- **Erasure** is the failure we set out to fix. **The erasure rate** is the number of true lesions the model failed to recover, divided by the number of true lesions, where a lesion counts as recovered if the prediction covers at least 10% of it. Lower is safer.
- **Hallucination** is the opposite failure: the reconstruction invents tumor where there is none. **The hallucination rate** is the number of predicted blobs that are fabricated, divided by the number of predicted blobs, where a blob counts as fabricated if less than 10% of it sits inside real tumor. Lower is safer, and it tends to move against erasure.
- **The floor** is what the tumor detector misses when handed the untouched original scan, with no degradation or reconstruction involved. It is large, so it is the denominator every erasure rate must be read against.
- **The baseline** is the distortion-optimal model: the standard approach, trained only to minimise average pixel error. **Ours** is the tumor-aware model: the same network and the same data, with errors inside the tumor weighted far more heavily.

The result

Sharpening a cheap blurry scan with AI can quietly delete a small tumor, because the model is graded on average pixel error and a small tumor is a tiny fraction of the pixels. We changed what the model is punished for, weighting errors inside the tumor far more heavily, and then measured how much tumor survives. On 70 held-out BraTS patients it loses far less tumor than the standard model, at the *same* segmentation quality.

What the tumor detector was given	Tumor missed	Caused by sharpening
The original scan, untouched	45.5%	—
Standard sharpening (baseline)	54.6%	+9.1 points
Tumor-aware sharpening (ours)	48.1%	+2.6 points

The detector already misses 45.5% on a perfect scan, so most of the middle column is nothing to do with sharpening. Sharpening adds 9.1 points on top of that floor and **our objective removes 71% of that damage**. 66 of 70 patients improve, 1 is worse, $p < 0.0001$. Quality is genuinely matched, not approximately: Dice 0.660 against 0.675, and brain-masked PSNR within 0.23 dB.

Is a rate near 50% bad? Split it by lesion size

Lesion size	How many	Floor	Baseline adds	Ours adds
Small, < 50 px	6,762 (71%)	65.1%	+13.5	+4.7
Medium, 50–200 px	1,005	10.0%	+6.4	+3.5
Large, > 200 px	1,723	0.5%	+0.7	+0.4

Read the bottom row first: a lesion above 200 px is missed 0.5% of the time on a perfect scan and 0.9% after our reconstruction, so substantial tumors are not vanishing. The overall rate is high because 71% of components are under 50 px and two thirds of those are missed even on the original

image. Small does not mean unimportant — a 30 px enhancing focus can be an early recurrence, and this is where the remaining work is — but it is also partly an artefact of counting a fragmenting rim as many equal lesions.

Three things that make it credible, and one that limits it

- **It appears only where the theory says it must.** The benefit is there for the small enhancing tumor and *vanishes* for the whole tumor region, nine times larger. That is the prediction: the objective is only misaligned when the structure is too small to matter to a pixel-error score.
- **The patient is the unit of analysis.** One tumor spans many slices and appears as a lesion in each, so those outcomes are correlated; pooling them fakes an interval about 1.5 times too tight. Resampling patients instead gives a mean paired difference of 6.5 points, 95% CI [5.2, 8.2].
- **The weakest link is the detector, not the sharpening.** It misses 45.5% of lesions on an untouched scan. A stronger downstream segmenter would buy more than a better loss function.
- **The limit: it is not free.** Protecting tumors makes the model readier to report one that is not there, and hallucination rises from 0.266 to 0.387. A second loss term recovers most of that cost for about half the benefit, so the tradeoff is a dial rather than a fixed price.

Provenance: enhancing-tumor checkpoint trained on 17,233 axial slices from 468 BraTS patients (Medical Segmentation Decathlon brain task), split by patient, measured on the 70-patient validation split. Raw outputs in results/; checkpoints at douyeszn/tumor-aware-sr on Hugging Face. The proposal's results table is from an earlier, smaller run, so its digits differ.